

Applying Conformal Prediction to a Deep Learning Model for Intracranial Hemorrhage Detection to Improve Trustworthiness

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See also commentary by Ngum and Filippi in this issue.

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Purpose: To apply conformal prediction to a deep learning (DL) model for intracranial hemorrhage (ICH) detection and evaluate model performance in detection as well as model accuracy in identifying challenging cases.

Materials and Methods: This was a retrospective (November–December 2017) study of 491 noncontrast head CT volumes from the CQ500 dataset, in which three senior radiologists annotated sections containing ICH. The dataset was split into definite and challenging (uncertain) subsets, in which challenging images were defined as those in which there was disagreement among readers. A DL model was trained on patients from the definite data (training dataset) to perform ICH localization and classification into five classes. To develop an uncertainty-aware DL model, 1546 sections of the definite data (calibration dataset) were used for Mondrian conformal prediction (MCP). The uncertainty-aware DL model was tested on 8401 definite and challenging sections to assess its ability to identify challenging sections. The difference in predictive performance (*P* value) and ability to identify challenging sections (accuracy) were reported.

Results: The study included 146 patients (mean age, 45.7 years $\pm$ 9.9 [SD]; 76 [52.1%] men, 70 [47.9%] women). After the MCP procedure, the model achieved an F1 score of 0.919 for localization and classification. Additionally, it correctly identified patients with challenging cases with 95.3% (143 of 150) accuracy. It did not incorrectly label any definite sections as challenging.

Conclusion: The uncertainty-aware MCP-augmented DL model achieved high performance in ICH detection and high accuracy in identifying challenging sections, suggesting its usefulness in automated ICH detection and potential to increase trustworthiness of DL models in radiology.

Supplemental material is available for this article.

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Intracranial hemorrhage (ICH) is a widespread and potentially life-threatening condition, affecting more than 50 000 adults annually in the United States (1). Deep learning (DL) has demonstrated promising results as an automated solution for ICH detection (2). However, DL models generally do not provide a confidence score for their predictions, making it difficult for both clinicians and researchers to trust them (3). Current state-of-the-art models perform well in hemorrhage detection but are not trustworthy in clinical settings because they may output incorrect predictions with high levels of confidence.

In general, a radiologic DL model provides predictions (diagnoses) in the form of numerical outputs for the possible diagnoses that were included in the training dataset. These outputs possess the mathematical properties of probability measures but are not representative of the true probability of a given outcome because they are uncalibrated (3–5). Calibration is the process by which a model's predictions are adjusted to more accurately approximate the underlying distribution of the population on which it will be applied. Conventionally, calibration maps a model's output to the observed frequency of an event. In the current study, however, calibration refers to the process of learning the test distribution from a small sample of the population called

the *calibration dataset*. Conformal prediction (ie, distribution-free uncertainty quantification) is a family of post hoc statistical methods, the premise of which is to generate prediction sets (ie, differential diagnoses) that are guaranteed to contain the true value (diagnosis) at a user-specified confidence level (6). Though conformal methods vary in nature and implementation, the general process begins by defining a heuristic notion of uncertainty for the DL model. Then, for every instance in the calibration dataset, this uncertainty metric is calculated and recorded. Finally, the uncertainty values obtained during calibration are then applied to some computation during inference to yield sets of predictions—instead of point predictions—for each test instance. The nature of this computation depends on the nature of the selected conformal method.

Mondrian conformal prediction (MCP) is a subset of conformal prediction that calculates the similarity of a given sample to each class for the problem at hand (6). A group is defined as the aggregation of model scores for a specific class across all calibration samples. When generating groups, a score for each class is obtained for every sample, regardless of its true label. MCP operates by ranking the calibration heuristic notion of uncertainty values across a given group, which allows for the calculation of conformal scores for predictions

Abbreviations

AUC = area under the receiver operating characteristic curve, DL = deep learning, FN = false negative, FP = false positive, ICH = intracranial hemorrhage, IoU = intersection over union, MCP = Mondrian conformal prediction, TN = true negative, TP = true positive

Summary

An uncertainty-aware deep learning model developed using Mondrian conformal prediction demonstrated high performance in intracranial hemorrhage detection on noncontrast head CT volumes and high accuracy in identifying challenging cases.

Key Points

- An uncertainty-aware deep learning model was trained to localize and classify five types of intracranial hemorrhage and demonstrated state-of-the-art performance ($F1 = 0.92$).
- The conformal prediction-based model identified patients with challenging intracranial hemorrhage cases with 95.3% accuracy.

Keywords

CT, Head and Neck, Brain, Brain Stem, Hemorrhage, Feature Detection, Diagnosis, Supervised Learning

on the test dataset. For each test sample, a conformal score is obtained for each group by dividing the would-be rank (ie, the insertion index) of the sample's heuristic notion of uncertainty value within a given group by the size of that group. This yields G conformal scores for each sample in the test set, where G is the number of groups (classes). This conformal score is then compared against a user-specified threshold, and groups for which the sample's score exceeds the threshold are included in its final prediction set.

In addition to uncalibrated models, another source of mistrust in DL models is poor handling of out-of-domain cases. Traditional training methods restrict the apparent knowledge of a DL model to the limits of its training domain (7), such that as inputs trend away from being in domain, the model's predictions are increasingly likely to be incorrect. Artifacts like patient motion or implanted devices can also substantially alter images, causing them to be out of domain. The out-of-domain problem is exacerbated when a model's outputs are not calibrated.

This study used MCP as a method to improve trustworthiness in DL models. We trained an object detection algorithm to classify and localize ICH in noncontrast head CT sections. We evaluated the performance of the MCP-augmented model in detection of five classes of ICH, including challenging sections. Challenging sections were defined as those in which radiologists disagreed on whether ICH was present.

Materials and Methods

Data Collection

This was a retrospective study of 491 anonymous, noncontrast head CT volumes collected at various centers in New Delhi, India, in 2017 and compiled as the publicly available CQ500 dataset (8). The multi-institutional 2019 RSNA Brain CT Hemorrhage Challenge dataset (21 784 total patients; 1999–2014 and 2018) was used to externally test the performance of our DL model (9). See Appendix S1 and S2 for more information about the datasets used in this study. Because both datasets were anonymized and publicly available, institutional review board approval was not required.

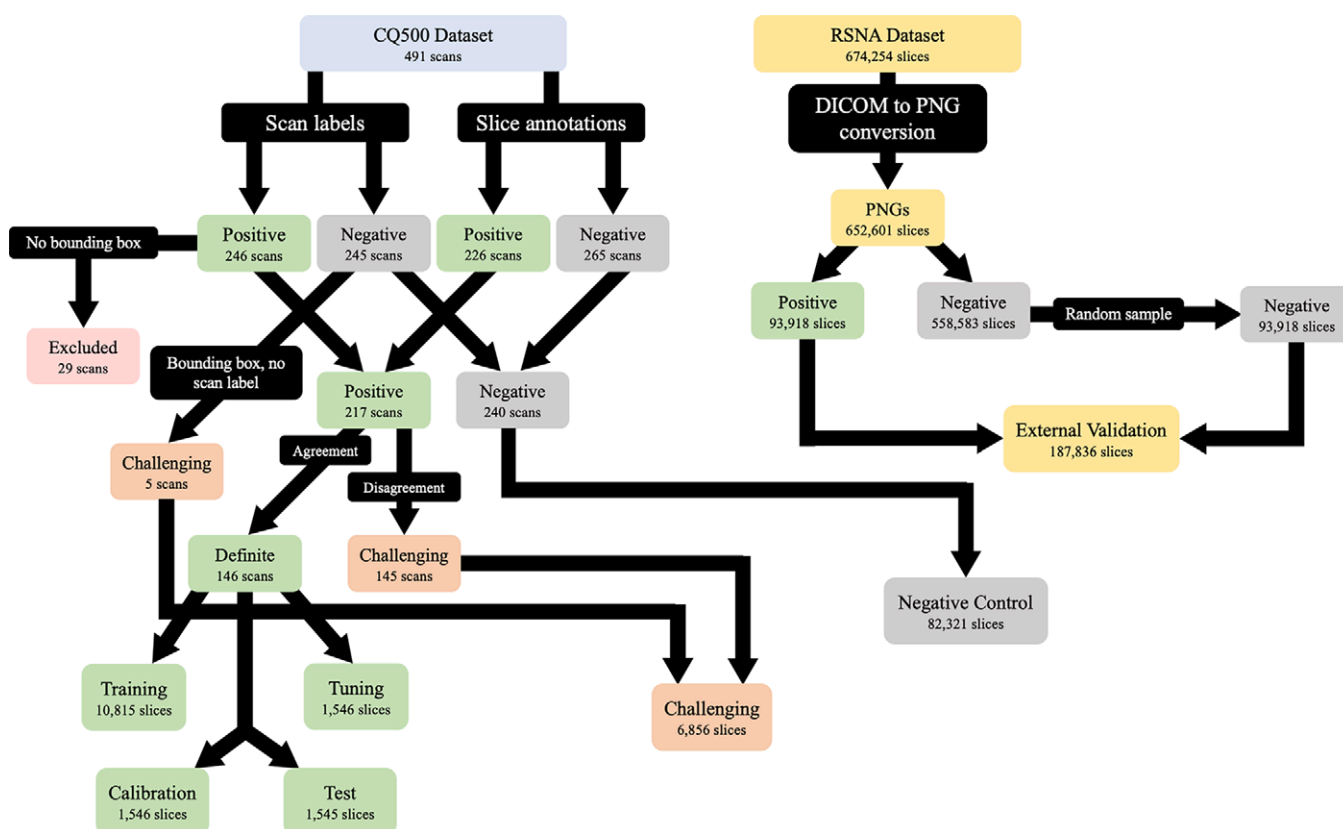


Figure 1: Depiction of data inclusion, exclusion, and splitting. RSNA = Radiological Society of North America, DICOM = Digital Imaging and Communications in Medicine, PNG = portable network graphic.

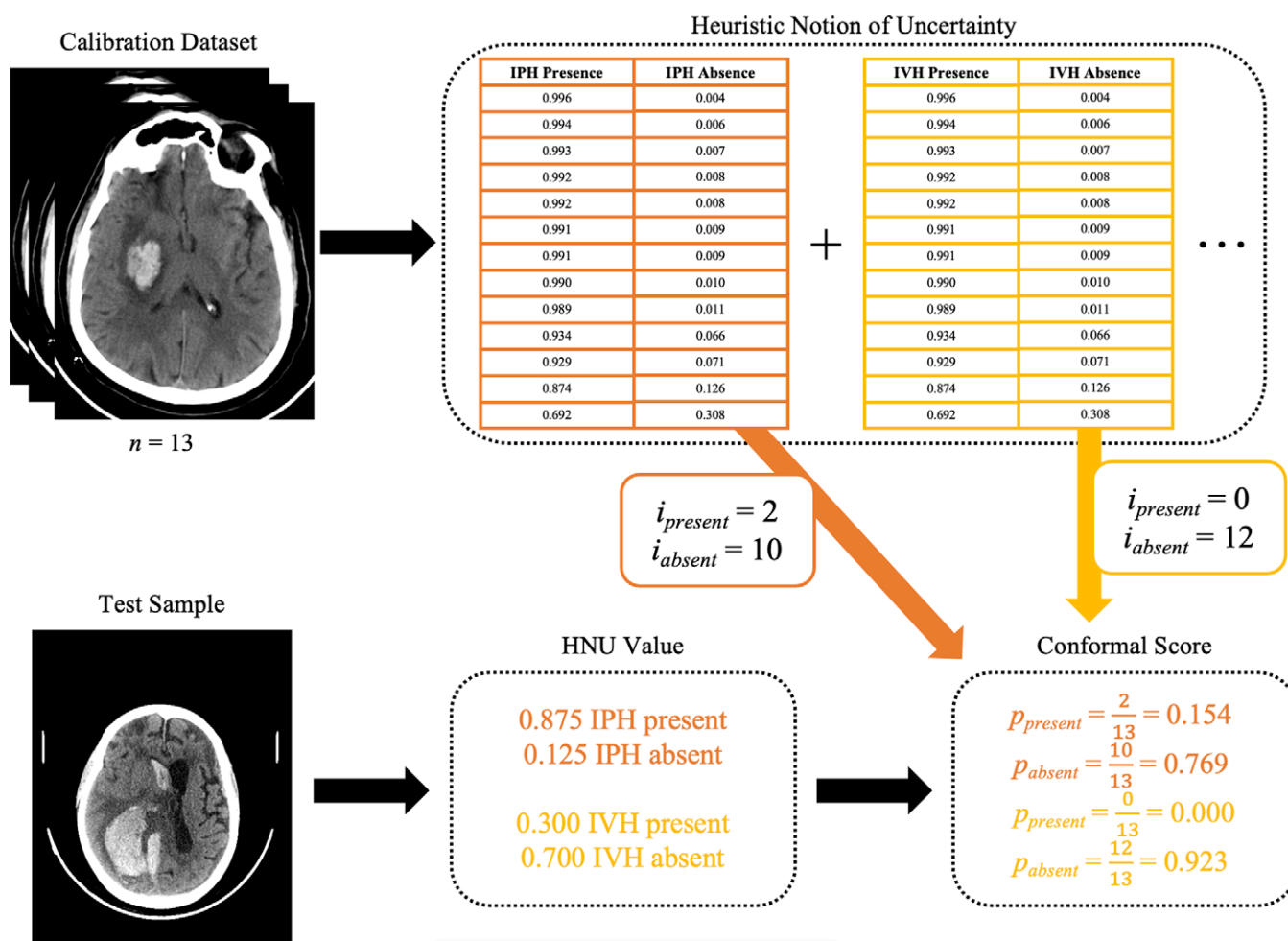


Figure 2: High-level overview of Mondrian conformal prediction as it is applied in this study. Note that heuristic notion of uncertainty (HNU) lists are zero indexed. IPH = intraparenchymal hemorrhage, IVH = intraventricular hemorrhage, i = insertion index in sorted value array, P = conformal score.

Dataset Organization

In the CQ500 dataset, 29 volumes—for which all three readers (with 8, 12, and 20 years of experience in cranial CT interpretation, respectively) indicated that none of the five hemorrhage classes (intracranial, intraparenchymal, intraventricular, subdural, epidural) were present—were excluded and used as an evaluative control group. After narrowing the dataset to contain only volumes positive for at least one class of hemorrhage, the volumes were split into definite and challenging subsets. The definite dataset (146 patients, no patients with multiple volumes) was defined to contain volumes with labels in which all readers unanimously agreed. Any scan for which there was not unanimity among readers for the five hemorrhage classes was placed into the challenging dataset (150 patients, no patients with multiple volumes). The definite dataset was then further divided into training, tuning, calibration, and test subsets with a 70, 10, 10, and 10 split (10), assuring that images were grouped by their series and patient identifiers to prevent information leakage between subsets (10). Sections were also stratified by their consensus labels from the three readers for all five hemorrhage types to maintain a class distribution as close to the true distribution as possible. The data inclusion, exclusion, and splitting process is depicted in Figure 1.

The external test dataset was split into positive (hemorrhagic) and negative (nonhemorrhagic) subsets. Because the negative subset was substantially larger than the positive subset, and because a negative control group from the CQ500 dataset already existed, a random sample of the negative subset was used during external testing. The sample size was equal to the total number of positive sections.

See Appendix S3 and S4 for additional remarks about the data organization process.

Image Processing

Windowing was applied to all volumes in the final datasets at a window level of 50 HU and a window width of 80 HU to enhance hemorrhage prominence (11). The windowed volumes were then converted from their original Digital Imaging and Communications in Medicine format into two-dimensional portable network graphic images of 512×512 pixels. A total of 21 653 of 674 254 images in the RSNA dataset were missing Digital Imaging and Communications in Medicine headers and could not be converted to portable network graphic format, so these images were excluded from the final external test dataset. During training, the DL pipeline applied standard augmentations to reduce the effects of overfitting (12). Augmentations were not applied during

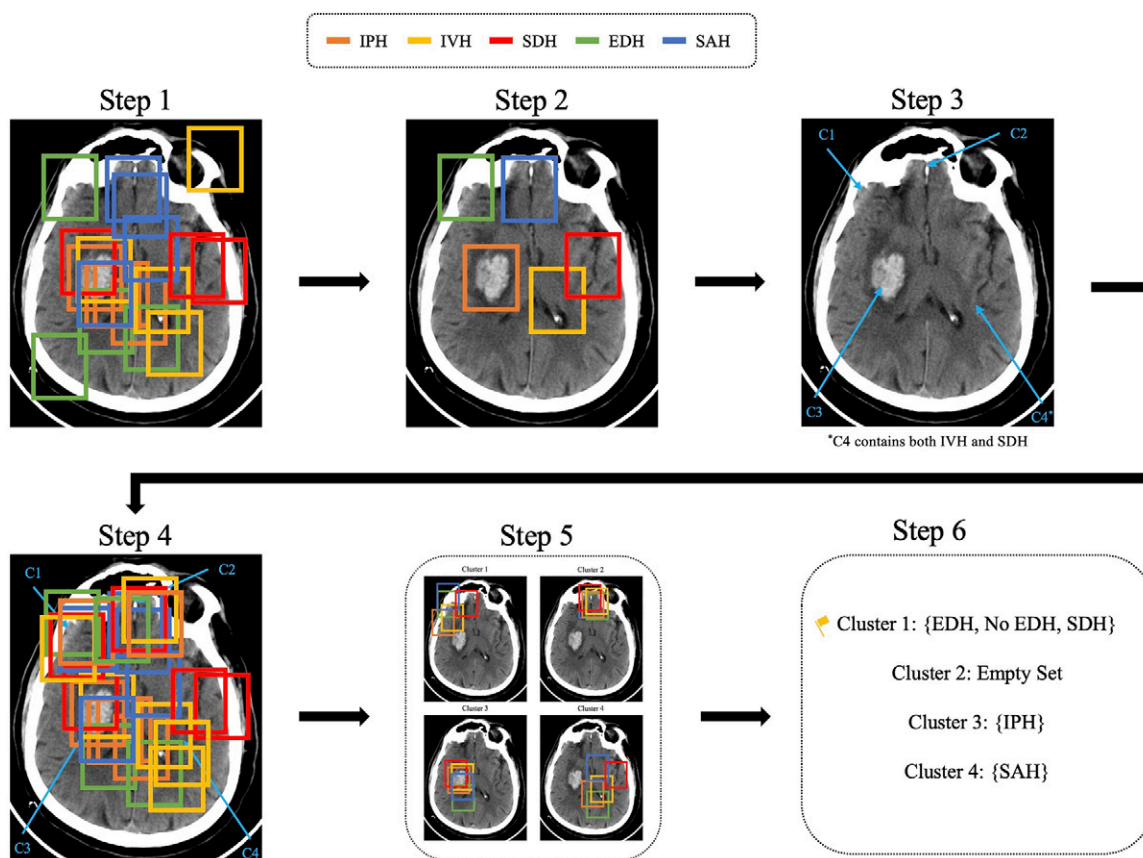


Figure 3: Algorithm for clustering predictions with intersection over union (IoU) thresholding and nonmaximum suppression. Step 1 allows all predictions. Step 2 selects the maximum confidence predictions for each hemorrhage class. Step 3 creates clusters based on IoU calculations between each prediction. Step 4 reinspects the filtered predictions to obtain unfiltered sets for each cluster. Step 5 selects the maximum confidence predictions per cluster per class. Step 6 obtains the P values for each prediction on each cluster. Note that cluster 1 containing epidural hemorrhage (EDH) and no EDH is an indication of uncertainty. C1 = cluster 1, C2 = cluster 2, C3 = cluster 3, C4 = cluster 4, IoU = intersection over union, IPH = intraparenchymal hemorrhage, IVH = intraventricular hemorrhage, SAH = subarachnoid hemorrhage, SDH = subdural hemorrhage.

other phases of the study. Additional information regarding the augmentations used can be found in Table S1.

Model Development

Ultralytics's You Only Look Once computer vision model, version 8, was selected as the DL tool for ICH detection because it is well regarded as state of the art in object detection (13). The You Only Look Once pipeline outputs both the model's confidence score for a bounding box containing an object of interest as well as its confidence score for a detected object belonging to the given class. The final confidence score for a given prediction is determined by a nonlinear combination of these two confidence values, whose weighting is optimized as an auxiliary component of the cost function during training. The model was trained for 300 epochs with a batch size of six using the AdamW optimization algorithm with an initial learning rate of 0.01 and You Only Look Once's cosine annealing learning rate scheduler (14). At the end of each training epoch, the model was applied to the tuning dataset to benchmark its performance on data it had not seen during training. Additional training parameters are listed in Table S2.

Model Calibration

When the model completed training, inference was performed on the calibration dataset. The model was allowed

to predict up to 10 000 instances of hemorrhage per section because to populate class-wise Mondrian groups, a bounding box and confidence score for each hemorrhage class was needed for every section. These predictions were then filtered by identifying the highest confidence bounding box for each hemorrhage class, resulting in five confidence scores—one for each hemorrhage class—for each calibration sample. Five Mondrian groups were then formed by sorting the calibration confidence scores across all samples for each hemorrhage class. An additional five groups were formed for the absence of each hemorrhage class by sorting the additive complements of all calibration confidence scores within each hemorrhage class. Figure 2 demonstrates the flow of MCP from calibration to inference.

Model Performance Evaluation

After generating Mondrian groups, the model was evaluated on the test, challenging, negative control, and RSNA external test datasets. During inference for each dataset, the model was once again allowed to predict up to 10 000 instances of hemorrhage per section (Fig 3, Step 1). The highest confidence bounding boxes for each hemorrhage class were then selected, yielding five bounding boxes—each of a different class label—for each sample (Fig 3, Step 2). Next, clusters were created by calculating the intersection over union (IoU)

	Matrix A Localization	Matrix B Classification and Localization	Matrix C Exact Classification and Localization
TP	There exists a prediction whose IoU with a GT is greater than or equal to the given threshold.	There exists a prediction whose IoU with a GT is greater than or equal to the given threshold and whose label is not equal to the GT label.	There exists a prediction whose IoU with a GT is greater than or equal to the given threshold and whose label is equal to the GT label, and there does not exist a prediction whose IoU with that GT is greater than or equal to the given threshold and whose label is not equal to the GT label.
FP	There exists a prediction whose IoU with all GTs is less than the given threshold.	i. There exists a prediction whose IoU with all GTs is less than the given threshold. ii. There exists a prediction whose IoU with a GT greater than or equal to the given threshold and whose label is not equal to the GT label, and there does not exist a prediction whose IoU with that GT is greater than or equal to the given threshold and whose label is equal to the GT label.	i. There exists a prediction whose IoU with all GTs is less than the given threshold. ii. There exists a prediction whose IoU with a GT is greater than or equal to the given threshold and whose label is not equal to the GT label, there does not exist any other prediction whose IoU with that GT greater than or equal to the given threshold.
TN	There does not exist a prediction whose IoU with all GTs is less than the given threshold.	i. There does not exist a prediction whose IoU with all GTs is less than the given threshold. ii. There exists a GT for which there does not exist a prediction whose IoU with the GT is greater than or equal to the given threshold and whose label is not equal to the GT label.	N/A
FN	There exists a GT whose IoU with all predictions is less than the given threshold.	i. There exists a GT whose IoU with all predictions is less than the given threshold. ii. There exists GT for which there does not exist a prediction whose IoU is greater than or equal to the given threshold and whose label is not equal to the GT label.	N/A

Figure 4: Textual descriptions of true positive (TP), false positive (FP), true negative (TN), and false negative (FN) for the three confusion matrices used during model evaluation. GT = ground truth, IoU = intersection over union, N/A = not applicable, RS = reference standard.

between each bounding box and all other bounding boxes (Fig 3, Step 3). A cluster was defined by any bounding box whose IoU with all other bounding boxes was less than the given threshold. Then, each cluster was filled by reinspecting all initial predictions, excluding those that had already been selected, and including any bounding box whose IoU with the cluster-delimiting bounding box was greater than or equal to the given IoU threshold (Fig 3, Step 4). Pen-ultimately, each cluster was condensed to contain only five bounding boxes by retaining the cluster-delimiting bounding box and the highest confidence bounding box for the remaining four hemorrhage classes (Fig 3, Step 5). Finally, conformal scores were calculated for each bounding box, and conformalized prediction sets were generated for each cluster (Fig 3, Step 6).

This algorithm relied on two threshold values: IoU between a given bounding box and a cluster-delimiting bounding box was used as a threshold so that users could specify the degree of strictness by which they form clusters, and the conformal score threshold of a given bounding box in a populated cluster allowed users to specify the allowable degree of uncertainty in their final prediction sets. As such, an optimization process was conducted with the calibration dataset to determine which thresholds to apply during evaluation on the test, challenging, negative, and external test datasets. Three confusion matrices were defined. Each matrix tested different criteria for a prediction to be counted as true positive (TP), false positive (FP), true negative (TN), or false negative (FN). Precise textual definitions of TP, FP, TN, and FN for each confusion matrix can be found in Figure 4.

During optimization, IoU and conformal score thresholds were varied between 0.00 and 1.00 at intervals of 0.05. The clustering algorithm described above and depicted in Figure 3 was then applied using each combination of IoU and conformal score thresholds, yielding 441 conformalized prediction sets (differential diagnoses) for each calibration instance. Then, for matrices A and B, the IoU and conformal score thresholds that yielded the maximum area under the receiver operating characteristic curve (AUC) were selected. For matrix C, because only TP and FP were defined, the threshold values that yielded the maximum positive predictive value were selected. To avoid confusion in the Results section, note that the threshold values obtained during optimization for both matrix A and matrix B were identical.

These threshold values were then applied to the test, challenging, negative, and external test datasets. For the test and challenging datasets, all four counts (TP, FP, TN, FN) as well as sensitivity, specificity, positive predictive value, negative predictive value, and the F1 score were reported for the thresholds obtained from matrices A and B. For matrix C, TN and FN were not well defined, so only positive predictive value was reported. For the control group, only the TN and FP counts and specificity were reported. Other metrics were not defined because this dataset contained only nonhemorrhagic sections. No bounding boxes were available for the external validation dataset, so the evaluation was agnostic to localization performance and assessed only the model's classification ability. As such, the definitions of TP, FP, TN, and FN were adjusted to yield meaningful results (see Fig 5). For classification, sensitivity, specificity, positive predictive value,

	Classification	Exact Classification
TP	There exists a prediction in the prediction set whose label is equal to the GT label.	There exist only predictions in the prediction set whose labels are also GT labels, and the length of the prediction set is exactly the length of the set of GTs.
FP	There exists a prediction in the prediction set whose label is not equal to any GT label.	There exist only predictions in the prediction set whose labels are not equal to the GT labels, and the length of the prediction set is greater than or equal to the length of the set of GTs.
TN	There exists no label in the prediction set that does not also exist as a GT label.	Both the GT and prediction are sets of length zero.
FN	There exists a GT label that is not equal to the label of any prediction in the prediction set.	N/A

Figure 5: Textual descriptions of true positive (TP), false positive (FP), true negative (TN), and false negative (FN) for the two confusion matrices used during external validation of the model. GT = ground truth, N/A = not applicable.

Class	Training	Tuning	Calibration	Test	Challenging
IPH	4743	664	664	683	732
IVH	1415	219	208	187	407
SDH	3183	432	466	451	4251
EDH	273	40	40	39	174
SAH	5237	754	765	745	3557
Combined	14 581	2109	2143	2105	9121

Note.—Data are numbers of hemorrhages. A single section can contain multiple hemorrhages. EDH = epidural hemorrhage, IPH = intraparenchymal hemorrhage, IVH = intraventricular hemorrhage, SAH = subarachnoid hemorrhage, SDH = subdural hemorrhage.

negative predictive value, and the F1 score were reported. For exact classification, because FN was not well defined, only specificity and positive predictive value were reported.

In addition to the metrics described above, mean average precision was reported for all datasets except the control group because it contained no positive samples and the external test dataset because bounding boxes were unavailable. Mean average precision is a standard metric in object detection, and though it does not paint a full picture of a model's performance, it can offer insight into its positive predictive ability (14,15). To aggregate TP, FP, TN, and FN

values across each hemorrhage class, the value of each metric was summed across the five classes. The result was then divided by the total number of hemorrhage instances. This allowed for more granular insights into the model's performance on each hemorrhage, even if there were multiple hemorrhages in a given section.

Trustworthiness Evaluation

This study also evaluated the performance of MCP as a mechanism for trustworthiness in DL. The accuracy of the model in identifying challenging sections was reported; a sample was predicted as challenging if the final prediction set for any cluster within the sample contained either of the following: both the presence and absence of one hemorrhage class or the presence of more than one hemorrhage class. The number of samples in the definite test dataset that the model incorrectly labeled as challenging was also reported to assess the model's ability to distinguish between definite and challenging sections.

Statistical Analysis

All calculations were performed using the NumPy (version 1.24.4) and scikit-learn (version 1.2.2) libraries in Python (version 3.8.16; Python Software Foundation). Welch *t* test to compare two sample means was used to measure the significance of the difference in model performance on challenging

versus definite sections by comparing the means of the per-instance average precision at an IoU threshold of 0.95.

Code Availability

A GitHub repository (<https://github.com/c-gamble/radiology-mcp-toolkit>) is available, containing a deployable toolkit for trustworthiness in radiologic DL. It implements conformal prediction as proposed in this article and includes downloadable model weights for hemorrhage detection as well as a sample inference script, detailed usage instructions, and thorough explanations of the methods used.

Results

Dataset Characteristics

The definite dataset included 146 patients (mean age, 45.7 years $\pm$ 10.1 [SD]; 76 [52.1%] men, 70 [47.9%] women) and was split into 10 815 training sections, 1546 tuning sections, 1546 calibration sections, and 1545 test sections. The challenging dataset included 150 patients (mean age, 51.1 years $\pm$ 9.6; 77 [51.3%] women, 73 [48.7%] men) and comprised 6856 sections. Frequencies for each hemorrhage class within each dataset are shown in Table 1.

Of the 652 601 sections from the RSNA dataset (population statistics unavailable) successfully converted to portable network graphic images, there were 32 564 instances of intraparenchymal hemorrhage, 23 766 instances of intraventricular hemorrhage, 42 496 instances of subdural hemorrhage, 2761 instances of epidural hemorrhage, and 32 122 instances of subarachnoid hemorrhage.

Model Development Cost

Training the model for 300 epochs required 10.867 hours on three NVIDIA A100, SXM4, 80 GB cards. Inference was completed on a single NVIDIA card and required approximately 40 hours (Appendix S5).

Model Performance: Hemorrhage Detection

Table 2 displays the mean average precision values at an IoU threshold of 0.95 (mean average precision, 0.95) for each hemorrhage class for each dataset.

For the test and challenging datasets, confusion matrices with IoU and conformal score thresholds optimized on the test dataset are shown in Table 3.

Table 2: Mean Average Precision Values at IoU Threshold of 0.95 of All Hemorrhage Classes for Each Dataset

Class	Training	Tuning	Calibration	Test	Challenging
IPH	0.979	0.861	0.876	0.873	0.0492
IVH	0.958	0.804	0.798	0.851	0.040
SDH	0.976	0.891	0.880	0.884	0.0538
EDH	0.964	0.889	0.877	0.867	0.0379
SAH	0.980	0.837	0.842	0.864	0.00659
Combined	0.972	0.856	0.855	0.868	0.0375

Note.—EDH = epidural hemorrhage, IoU = intersection over union, IPH = intraparenchymal hemorrhage, IVH = intraventricular hemorrhage, SAH = subarachnoid hemorrhage, SDH = subdural hemorrhage.

Table 3: Confusion Matrices and Derived Metrics for Test and Challenging Datasets

Metric	Matrix A	Matrix B	Matrix C
Test dataset			
TP	1952	1950	347
FP	187	187	30
TN	1358	1358	...
FN	153	155	...
Sensitivity (%)	92.7 (1952/2105)	92.6 (1950/2105)	...
Specificity (%)	87.9 (1358/1545)	87.9 (1358/1545)	...
PPV (%)	91.3 (1952/2139)	91.2 (1950/2137)	92.0 (347/377)
NPV (%)	89.9 (1358/1511)	89.8 (1358/1513)	...
Challenging dataset			
F1	0.920	0.919	...
TP	1325	1325	15
FP	2222	2222	107
TN	4634	4634	...
FN	7796	7796	...
Sensitivity (%)	14.5 (1325/9121)	14.5 (1325/9121)	...
Specificity (%)	67.6 (4634/6856)	67.6 (4634/6856)	...
PPV (%)	37.4 (1325/3547)	37.4 (1325/3547)	12.3 (15/122)
NPV (%)	37.3 (4634/12 430)	37.3 (4634/12 430)	...
F1	0.209	0.209	...

Note.—Data in parentheses are numbers of correct hemorrhages and numbers of total hemorrhages. Test metrics were calculated by optimizing the intersection over union (IoU) and conformal score thresholds for area under the receiver operating characteristic curve (matrices A and B) and for positive predictive value (PPV; matrix C). Challenging metrics were calculated using the optimal thresholds as determined by the test dataset. FN = false negative, FP = false positive, NPV = negative predictive value, TN = true negative, TP = true positive.

Table 4 depicts the model's performance on the negative and external test datasets using the same optimal threshold values.

The model achieved an F1 score of 0.920 for ICH classification on the test dataset. Examples of TP, FP, TN, and FN are shown in Figure 6.

Model Performance: Identification of Challenging Cases

The model achieved an identification accuracy of 95.3% (143 of 150 patients) at the optimized IoU and conformal score thresholds. This accuracy represents the percentage of challenging sections that were correctly identified as challenging

Table 4: Performance Metrics for the Negative Control and External Test Datasets

Metric	Matrices A and B Thresholds	Matrix C Thresholds
Negative control		
TN	61 637	82 074
FP	20 684	247
Specificity (%)	74.9 (61 637/82 321)	99.7 (82 074/82 321)
External test (classification)		
TP	131 994	94 225
FP	60 088	839
TN	127 748	186 997
FN	55 842	93 611
Sensitivity (%)	70.3 (131 994/187 836)	50.2 (94 225/187 836)
Specificity (%)	68.0 (127 748/187 836)	99.6 (186 997/187 836)
PPV (%)	68.7 (131 994/192 082)	99.1 (94 225/95 064)
NPV (%)	69.6 (127 748/183 590)	66.6 (186 997/280 608)
F1	0.695	0.666
External test (exact classification)		
TP	27 295	302
FP	43 934	668
TN	60 284	93 624
Specificity (%)	57.8 (60 284/104 218)	99.3 (93 624/94 292)
PPV (%)	38.3 (27 295/71 229)	31.1 (302/970)

Note.—Data in parentheses are numbers of correct hemorrhages and numbers of total hemorrhages. Metrics are reported for two sets of threshold values: the first set was determined by optimizing for area under the receiver operating characteristic curve (matrices A and B), and the second set was determined by optimizing for positive predictive value (PPV; matrix C). FN = false negative, FP = false positive, NPV = negative predictive value, TN = true negative, TP = true positive.

according to the definition presented in the Materials and Methods section. Note that these data are not reported in a table because they were recorded as a binary task on a single dataset—1 if the model correctly labeled a section as “challenging” according to the algorithm described in the Materials and Methods section and 0 otherwise. The model labeled none of the 1545 definite test dataset samples as challenging and performed markedly worse ($P < .001$) on challenging sections than definite ones (Table 2). Figure 7 shows an example of the model’s failure on a challenging case.

Discussion

This study aimed to propose a method for increasing trust in medical DL tools by demonstrating the efficacy of MCP for estimating certainty in ICH detection and supplying a deployable toolkit for applications in other tasks. The proposed approach demonstrated high accuracy (95.3%) in identifying patients with challenging cases, indicating that MCP is a promising method for enhancing trust in radiologic artificial intelligence systems. Furthermore, the GitHub repository and usage tutorial (<https://youtu.be/tccprwxq74Q>) released to accompany this work offers a practical, scalable implementation of MCP to further validate its performance.

DL-assisted ICH detection was first proposed in 2017 with Phong et al (16) demonstrating that a convolutional neural network could be trained to classify the presence or absence of ICH with 98.9% average accuracy. Recently, Tegnegg et al (17) assessed the performance differences between

strongly and weakly supervised models in ICH classification using the RSNA dataset as its primary dataset and the CQ500 dataset for external testing. They reported an AUC of 0.96 on their tuning subset of the RSNA dataset and an AUC of 0.95 on the CQ500 dataset. Additionally, Umapathy et al (18) obtained an F1 score of 0.97 on the CQ500 dataset with a convolutional ensemble trained on the RSNA dataset. As for conformal prediction, Vovk et al (19) introduced the underlying theory in 2005. More recently, Angelopoulos and Bates (6) released a comprehensive guide to conformal methods in modern DL, and Ding et al (20) proposed class-conditional conformal prediction, from which the methods used in this study were derived.

This work presents a DL solution to ICH detection that performs on par with state-of-the-art models and introduces MCP as a method of building trust with radiologic DL tools. We propose that if a DL tool can generate statistically guaranteed differential diagnoses and reliably identify when it encounters a challenging input, then physicians will be increasingly able to trust this tool and therefore better positioned to adopt it as an automated solution in their clinical or research workflow. Current approaches, including the default You Only Look Once pipeline, allow for a similar process of thresholding predictions by their confidence scores, but they do not calibrate these thresholds with the underlying data. In addition, they do not provide a statistical guarantee for the model’s predictions. The uncertainty-aware localization and classification model generates statistically guaranteed differential diagnoses, and the integration

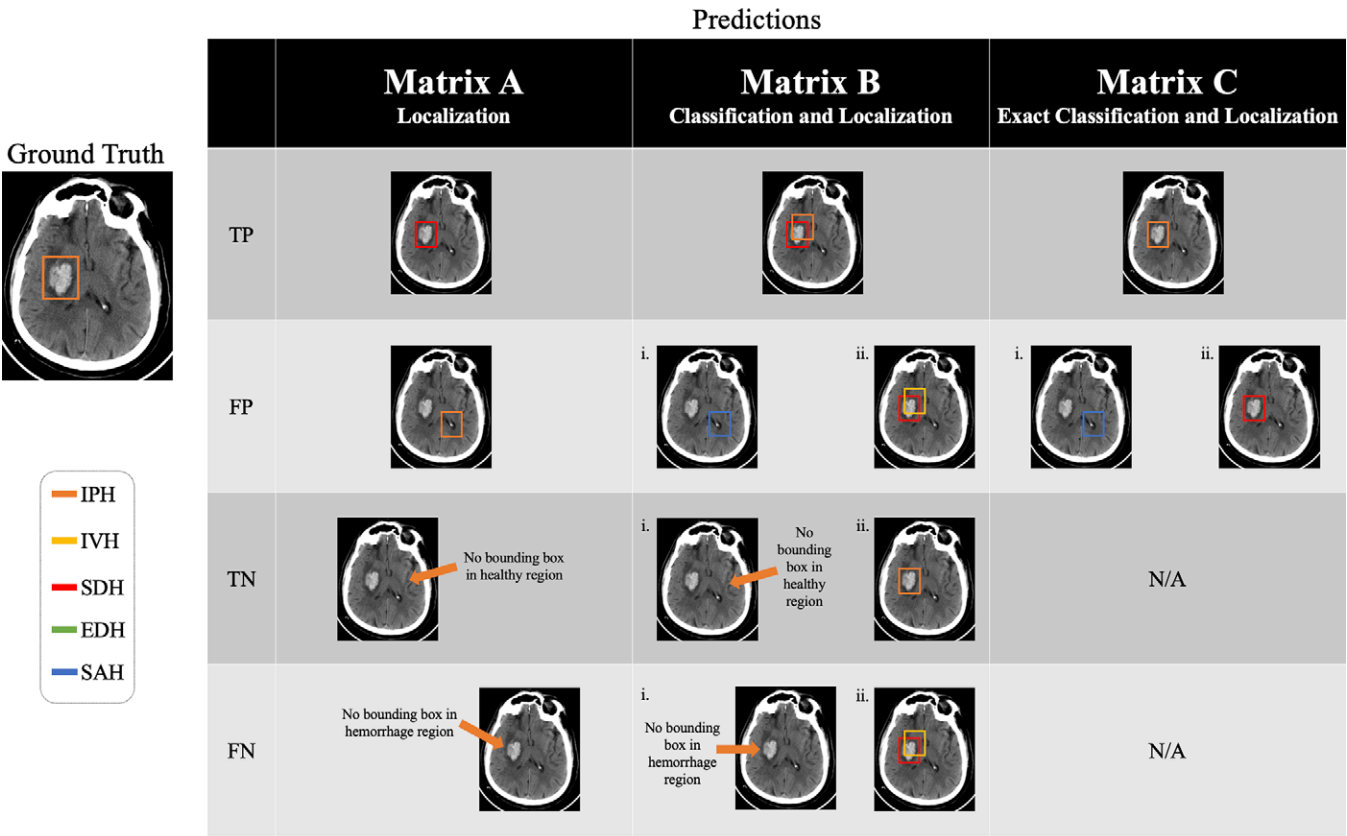


Figure 6: Visual depictions of examples of true positive (TP), false positive (FP), true negative (TN), and false negative (FN) for the three confusion matrices used during model evaluation. EDH = epidural hemorrhage, IPH = intraparenchymal hemorrhage, IVH = intraventricular hemorrhage, N/A = not applicable, SAH = subarachnoid hemorrhage, SDH = subdural hemorrhage.

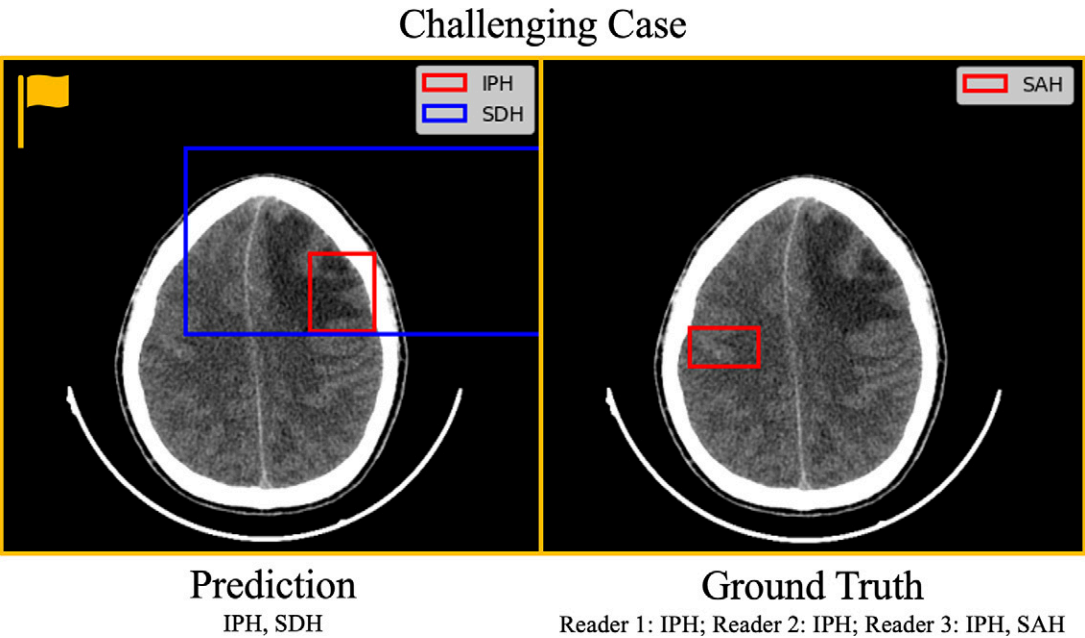


Figure 7: Example of challenging case prediction by the deep learning model compared with the reference standard. IPH = intraparenchymal hemorrhage, SAH = subarachnoid hemorrhage, SDH = subdural hemorrhage.

of MCP enables it to flag challenging cases with near-perfect accuracy for review by radiologists. Moreover, the DL architecture used in this solution makes it easily adaptable to a wide array of radiologic and medical tasks, especially considering that the IoU and conformal score thresholds can be optimized uniquely for each task to which this pipeline is applied. This opens the

door for further investigation into the critical frontier of trust in DL. The publicized code and deployment instructions further encourage the continued research and development of MCP as a mechanism for trust in radiologic DL.

This study had limitations. The DL model operates on two-dimensional CT sections, so it is not immediately

compatible with current picture archiving and communication systems. Given the current state of radiologic DL, however, this improvement is certainly feasible without significant modification to the MCP backbone of the trust component. Additionally, this study did not consider potential biases in the data or model.

To deem a DL algorithm trustworthy, it must be able to reliably communicate when it encounters cases about which it is uncertain. This study not only demonstrated that an MCP-based approach can yield near-perfect accuracy in the identification of challenging cases but also that a trustworthy DL model achieves performance metrics on par with state-of-the-art models. Further efforts in trustworthy radiologic DL models will include increasing practicality by expanding the application of MCP to three-dimensional models as well as continuous external testing to account for generalizability. Trustworthy DL is the necessary next step toward the feasible clinical adoption of artificial intelligence tools in radiology.

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